

Hardware & Firmware Integration

Medical Early Warning & Patient Monitoring Platform
Portfolio Reference Document | Prepared by Christopher Anthony Agustin | Senior Project Manager

1. Purpose

This document summarizes the hardware and firmware workstream and the coordination required to integrate wearable sensing, device electronics, embedded software, communication, data acquisition, and validation activities.

2. System Overview

Subsystem	Primary Responsibility	Integration Consideration
Optical Sensors	Capture PPG signals using RED / IR / GREEN emitters and photodiode sensing	Sensor geometry and signal quality affect measurement performance
Microcontroller	Control sensors, acquire signals, process data, and manage device functions	Firmware behavior must match hardware capabilities and timing
Power Management	Battery, charging, and power optimization	Power constraints influence sensing and device operating behavior
Connectivity	BLE communication and data transfer	Reliable transmission must align with gateway / application interfaces
Wearable Enclosure	Mechanical protection and sensor placement	Physical design affects sensor contact, usability, and manufacturing

3. Hardware / Firmware Delivery Flow

Stage	Key Activities	PM Control Point
Hardware Design	PCB, sensor, enclosure and component design	Scope, interfaces, dependencies
Firmware Development	Sensor control, acquisition, processing and communication	Backlog, sequencing, technical dependencies
Integration	Bring hardware and firmware together and verify interfaces	Defect tracking, ownership, retest
Calibration / Optimization	Assess signal quality and tune measurement performance	Coordinate engineering evidence and decisions
Verification & Validation	Performance, integration and validation testing	Readiness, traceability, issue closure
Production Readiness	Finalize engineering outputs for build and manufacturing	Handover, readiness and follow-up actions

4. PPG & SpO2 Calibration Considerations

PPG measurement performance depends on the physical sensor configuration as well as firmware and algorithm behavior. The calibration process considers sensor geometry, LED / photodiode positioning, wavelength selection, reference measurements, and collection of raw PPG and accelerometer data.

Consideration	Delivery Impact
Sensor geometry	Changes in LED / photodiode placement may change measurement characteristics and calibration needs.
Signal acquisition	Raw RED, IR, GREEN PPG and accelerometer data support analysis and calibration.
Reference measurements	Reference SpO2 and pulse-rate measurements provide comparison data for performance assessment.
Calibration	Parameters are derived for the specific sensor configuration and target measurement domain.

5. PM Coordination Responsibilities

- Coordinate hardware, firmware, software, quality, validation, and manufacturing dependencies.

- Surface technical blockers early and manage their effect on schedule, integration, and readiness.
- Ensure hardware and firmware changes are understood before downstream verification or validation.
- Coordinate evidence needed for performance assessment, calibration, integration, and release readiness.
- Maintain stakeholder alignment when technical decisions affect product scope or delivery timing.

6. Key Risks & Dependencies

Risk / Dependency	Typical Management Response
Sensor configuration changes	Reassess measurement behavior, calibration needs, and affected verification activities.
Hardware / firmware interface issues	Assign ownership, coordinate troubleshooting, retest, and update dependencies.
Signal quality / performance issues	Coordinate technical investigation, optimization, and validation evidence.
Manufacturing constraints	Align design decisions with production requirements and readiness milestones.

Portfolio Reference Document - based on project delivery context and technical evidence. It is not an official controlled company document and does not reproduce confidential source material.